

Laboratory 1:

Music Synthesis

Contents

1 Abstract

Each of the five senses are based on signals of different kinds. Musical signals can generally be heard, seen, and felt, and thus represent an excellent first place to explore signals and systems.

2 Objectives

In this assignment, you will use your expertise in MATLAB and your knowledge of sinusoids to synthesize the audio tones that make up the sequence of notes in a piece of music of your choice. You will use basic mathematical functions as the building blocks. In addition, you will explore how mathematical operations on these signals yield improvements in perceived quality and can be used to generate special effects.

You will have two weeks to complete this project.

- In the first week of lab, your goal (at a minimum) should be to implement a basic program that generates pure tones (Steps 1-6 in the Instructions).
- In the second week of lab, you should focus on implementing signal processing that improves the sound of your music (e.g., harmonics).

You may have to spend additional time outside of the two laboratory sessions to complete the project.

3 Pre-Laboratory Exercises

Before the lab, please read the following background information about music notation.

3.1 Music Crash Courses

Dr. Sienna M. Wood has created a website called [Music Crash Courses](#) to give information about both music theory and musicology. Several pages on her site will be helpful for this lab in terms of understanding how music can be represented on paper. Start by reading the following pages:

- [About Music Crash Courses](#) - it is always a good idea to find out what a web site is about before using it!
- [Introduction to Pitch](#) - one quick note is that the x axis in the graph should probably be time instead of frequency.
- [Introduction to Music Notation](#) - this gives a summary of information that you will read about later.
- [Notating and Naming Pitches](#) - this page will present a Grand Staff and will show what notes “look like” on it. Note that while there are seven “named” notes, there are actually twelve notes between one C and the next C - more on that soon. *Aside:* apparently, middle C is an explosive note!¹

¹Sorry, I'm not sorry.

- [Octave Numbering](#) - this page explains how to distinguish the same note in a different octave
- [Introduction to Intervals](#) - this page explains where the extra five notes come from (flats and sharps), what an *octave* is, and what some key frequencies are. Specifically, note that the A above middle C (called “middle A” or “A4”) is at 440 Hz and also that notes an octave below or above other notes are at half or double the frequency of the original.
- [Transposition](#) - for this page, mainly look at the accidentals in the score and then listen to how the music changes accordingly.
- [Major Scale](#) - this page begins a discussion of how the notation at the start of the Grand Staff might impact all the following notes. For the C major scale, the notes are unaltered.
- [Introduction to Keys](#) - this page starts looking at what happens if you have a piece of music that is **not** written in C major.
- [Relationships Between Keys - The Circle of Fifths](#) - this page shows the 24 possible keys (12 major, 12 minor) and the key accidentals. If the music is written in a key other than C major, the initial accidental for a note on the Grand Staff is applied to all similarly named notes regardless of octave. For instance, in G major, there is a sharp note for F3 and F5, but all F notes would be played as F sharp (unless another local accidental overrules that).
- [Rhythms in Notation](#) - this page describes the different symbols for different note lengths and different rest lengths. It also explains how dots and triplets change the duration.
- [Time Signature](#) - this page explains how the note timing for the song and for each measure is organized.

3.2 Improving Perceived Quality

- *Volume variations:* Typically, when a note is played, the volume rises quickly from zero and then decays over time, depending on how hard the key is struck and how long it is depressed. For a given tone, volume changes can be achieved by multiplying a sinusoid by another function called a windowing function. A decaying exponential is the simplest way to modulate the tone volume. You can also try concatenating different windowing functions (e.g., piecewise linear functions) to model the four segments of volume variation: Attack, Decay, Sustain, and Release (ADSR).
- *Overlapping tones:* Consider a beginning piano player's attempts at playing a simple tune. This student generally will quit pressing down one key before pressing down the next. A more advanced player would have some notes overlapping (either by using floor pedals or by pressing keys simultaneously). As the volume of one note is decaying another note is played. Mathematically this can be accomplished by allowing the time regions occupied by different sinusoids to overlap. This will yield a much smoother, less staccato-sounding piece.
- *Harmonics:* While single tones are represented by sinusoids, in real instruments the vibrations that occur when someone plays a particular "note" also include harmonic components. The frequency of the note is called the fundamental frequency f_0 ; the n^{th} harmonic is a sinusoid having a frequency equal to some integer n times the fundamental frequency: nf_0 . The amplitude, or power, of each harmonic component is generally less than that of the fundamental and decreases with increasing frequency. For each note, adding some lower-power harmonics makes the note sound "richer," whereas a single tone sounds very crisp. **NOTE:** adding in harmonics of equal or greater power will have undesired effects, such as increasing the perceived note fundamental, and may make the note sound harsh.
- *Instrument Synthesis:* You can modify your sinusoidal waveforms in many different ways to make the notes sound like particular instruments. For example, you can synthesize a bell using frequency modulation (FM). Specifically, your note will have the form:

$$x(t) = A(t) \cos(2\pi f_c t + I(t) \cos(2\pi f_m t))$$

where $A(t)$, the signal amplitude, and $I(t)$, both have an exponential decay form: $Ke^{-t/\tau}$. K is different for each, whereas τ is the same. The ratio of $f_c:f_m$ should be 2:1 for a bell. Other instruments can be synthesized using amplitude modulation (changing the amplitude envelope of the note).

- *Reverb:* Instruments are usually played in rooms that generate reverberations. Reverberations are reflections off of room surfaces, including walls, ceilings, floors and obstructions. In discrete time mathematics, you can view this as generating echoes of the original signal, with decreasing amplitude as the delay time increases. However, a major surface (e.g. a wall), may generate more power (amplitude) at specific delays. You can determine the delay time by dividing the distance sound travels by the speed of the sound (≈ 300 m/s at sea level). A model for generating an echo is given in the next section.

3.3 A Mathematical Model of the Echo Process

If you want to improve the sound of your score using reverb, you will need to create a function in MATLAB to generate an echo. A simple analytical model of an echo $s_e(t)$ of a signal $s(t)$ is $s_e(t) = \alpha(t - T)s(t - T)$, where $\alpha(t)$ attenuates the volume over time and T is the time delay introduced by reflection. In this treatment, we will assume that $\alpha(t) \in [0, 1)$ and $T \geq 0$. This implies that the echo's strength is weaker than the true signal and that the echo always arrives at a time later than the true signal. The signal $r(t)$ that you hear is formed by a simple addition of the original signal and its echo. This process is illustrated in the figure below for the simple case where $\alpha(t)$ is a constant. In this figure, $s(t)$ is the transmitted signal, $s_e(t)$ is that signal's echo, and $r(t)$ is the signal received by a listener. Thus in general, $r(t) = s(t) + s_e(t)$.

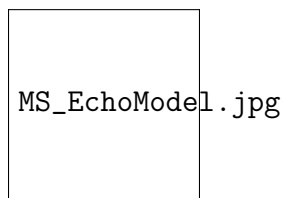


Figure 1. Echo Model

3.4 Pre-Laboratory Assignment

Given the following musical score (written in G-Major, as indicated by the key signature showing that all F notes should be played sharp):



“Mary Had a Little Lamb” - Lyrics: Sarah Josepha Hale, Music: Lowell Mason. Image from [Wikimedia Commons](#)

Consider:

1. The clef being shown,
2. The time signature,
3. The number of measures in the song, and
4. The pitch (including octave) and duration for each of the four times the word “lamb” appears

For an example of the latter, the first note (above “Ma”) would be an octave 4 quarter note B.

Pre-lab Deliverable (1/??): Complete the “ECE 280 Lab 1 Pre-lab Assignment” quiz on Canvas. You’ll have 5 attempts, and the maximum score of these 5 attempts will be kept.